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## Validation

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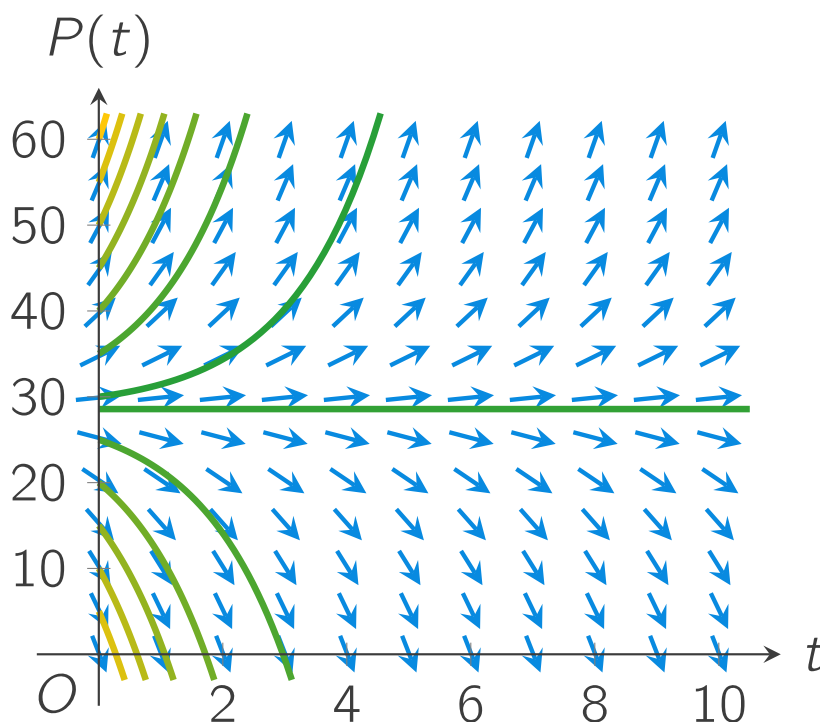
In the previous sections you have seen the differential equation

$$\frac{dP}{dt} = 0.7P(t) - 20,$$

with the initial condition  $P(0) = 30$ . The phase line for this differential equation is



and the direction field with some possible solutions is like this:



Now you will validate these solutions. Do these solutions represent realistic solutions to our problem?

### Exercise A

## Question

Are the solutions we have obtained realistic?



Answer



Reflection



Results

### Correct Answer

No

The solutions are not realistic, because if you start with 30 rainbowfish, the population grows within four days to more than a sixty rainbowfish. This number keeps increasing, which means at some time the aquarium will be really full (and might even break). Did you ever see this happen?

#### Your initial answer: No (Option 2)

👤 "Still the solution is not realistic since the solution is not always bounded from above with some initial values."

#### Your final answer: No (Option 2)

👤 "Still the solution is not realistic since the solution is not always bounded from above with some initial values."

See how your classmates answered below

### Class Breakdown

This is a look at how your classmates answered the question during the initial and final rounds.

### Answer Options

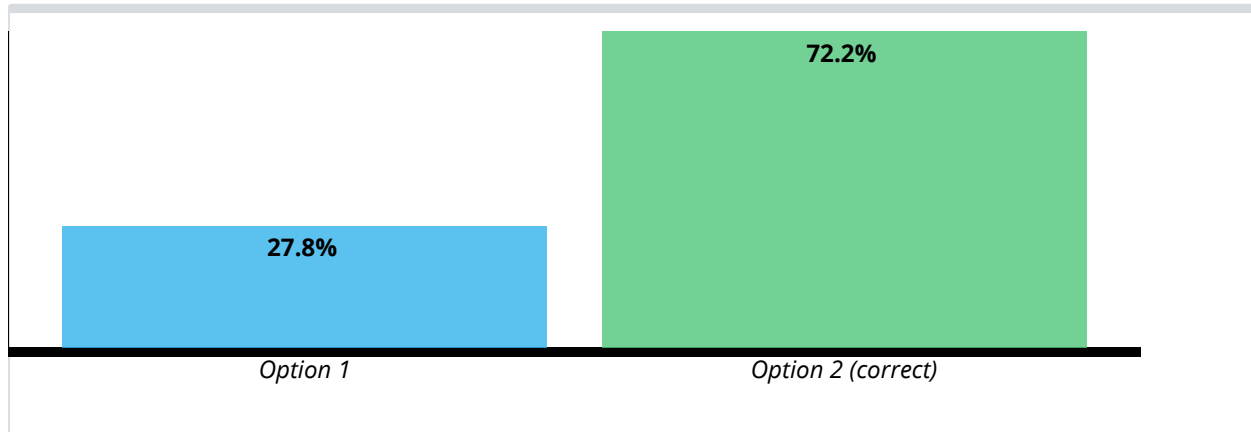
*Option 1*

Yes

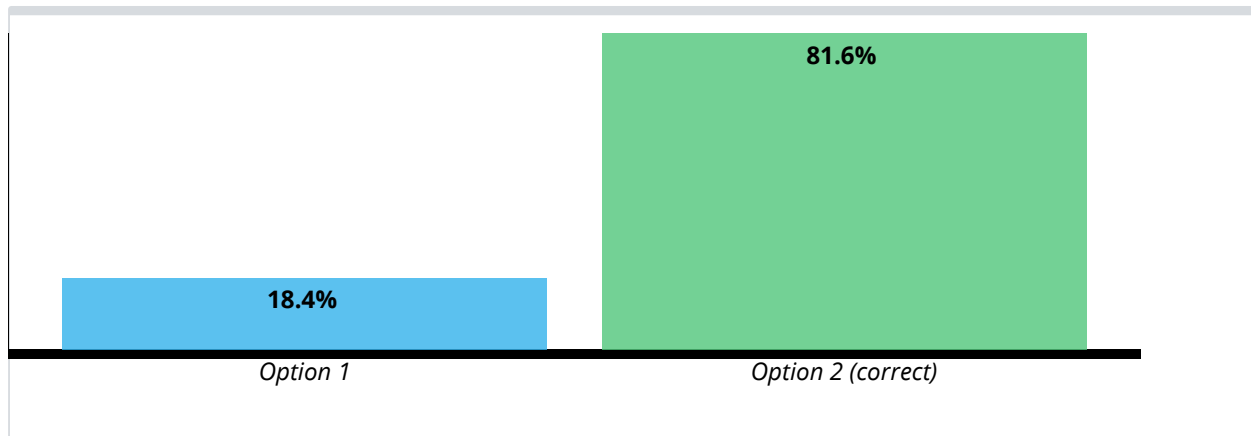
*Option 2 (correct)*

No

### Initial Answer Selection



### Final Answer Selection



In this section you have seen how to draw direction fields and sketch solutions for a differential equation. You also can draw a phase line and classify the type of the equilibrium points.

In the next discussion, we want you to think about why we introduced the distinction between stable and unstable equilibrium points.

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